RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019 – 2020 (Even Semester)

INNOVATIVE PRACTICE – Unit I

Degree, Semester & Branch: II Semester B.E. ECE A

Course Code & Title: EC8252 Electronic Devices

Name of the Faculty member: Dr.S.Periyanayagi

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ACTIVITY

Topic: Unit I Semiconductor Diode - Revision

Learning Outcome:

- The students will be able to explain the structure and characteristics parameters of PN diode

Activity Chosen: Visible Quiz

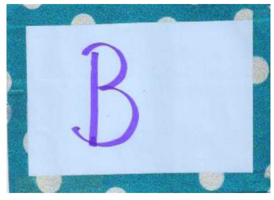
Justification:

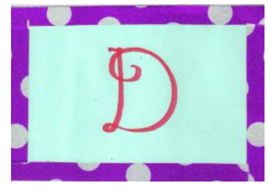
Electronic devices course is one of the fundamental subjects and the concept of PN diode plays a vital role in forthcoming courses in the higher semesters. In order to assess the understanding level of the students, visible quiz activity was conducted.

Time Allotted for the Activity: 10 Minutes

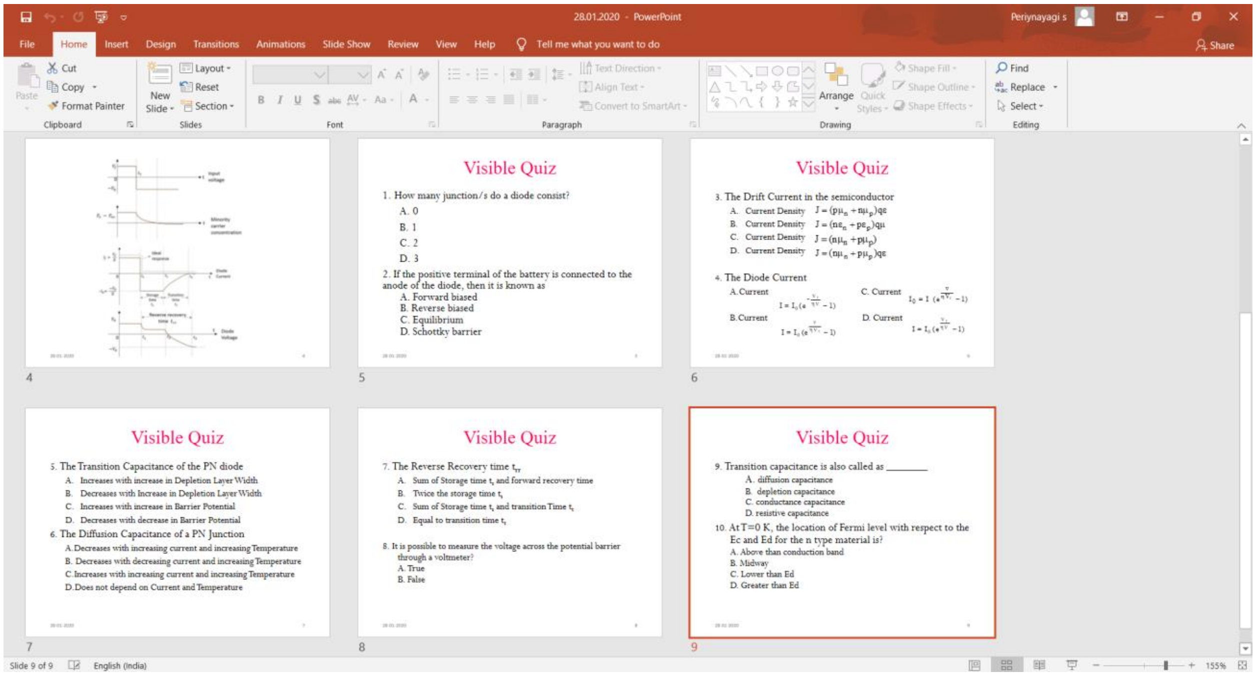
Details of the Implementation:

- Initially, 10 questions covering the important topic in the unit was taken and prepared in a power point presentation.
- The students were split into a group of four members (class strength: 40). 10 groups were formed and each member of the group were given with a card containing A, B, C, D as shown below.

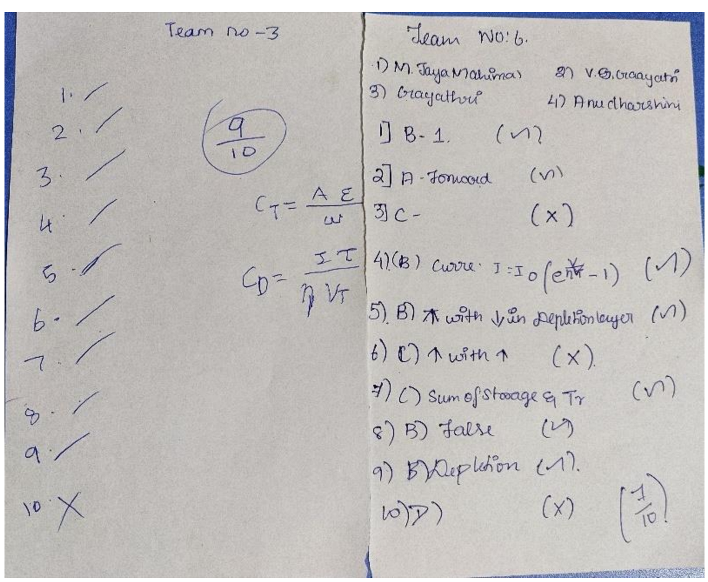




- Each group was provided with a paper to write down the answer for each question.
- Once the question is displayed one by one. The group member should raise the correct option card for the question.



- The other members will write the team answer in the paper given. Finally, the paper was interchanged with the group and finally discussed the correct answers.
- The highest scoring group was awarded with a pen as a token of appreciation.



Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was measured by the performance in the internal Assessment Test - I
- ✓ After completing the activity, the corrected paper with mistakes highlighted were given to the teams and discussed the correct answers.

Reflective Critique:

Benefits of conducting this activity:

- This activity helped to quickly recap the important fundament concepts related to PN Diode.
- Visible quiz is a team-based activity which supports collaborative learning through which the concepts will be made clear.

Challenges:

- This activity was planned for 10 minutes but it took 15 minutes for completing the discussion.
- The highest mark obtained by the group is 9/10.

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	1	1	-	-	-	-	2	-	3	-	2

CO1: The students will be able to Explain the structure, operations and characteristics of PN Junction diode

References:

1. Salivahanan S, Suresh Kumar N , Vallavaraj A, ‘ Electronic devices and circuits, 3 rd edition Tata Mcgraw Hill, 2008.
2. https://www.sanfoundry.com/electronic-devices-circuits-questions-answers-pn-junction-diode/

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019 – 2020 (Even Semester)

INNOVATIVE PRACTICE – Unit II

Degree, Semester & Branch: II Semester B.E. ECE A

Course Code & Title: EC8252 Electronic Devices

Name of the Faculty member: Dr.S.Periyanayagi

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ACTIVITY

Topic: h-parameter Model

Learning Outcome:

- The students will be able to apply the h parameter model while designing amplifiers.

Activity Chosen: Four Corner Four Question

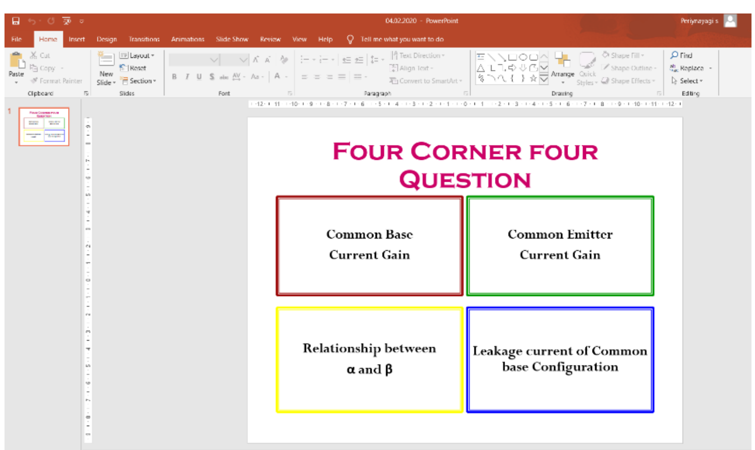
Justification:

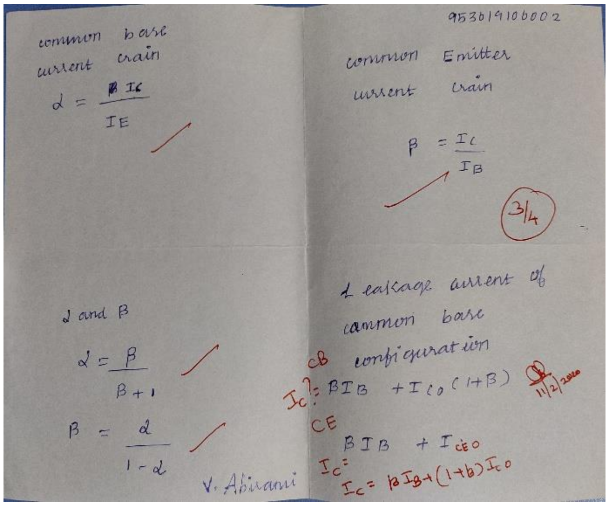
The concept of h parameter model is important and will be used in future courses like Electronic circuits I, Linear Integrated circuits. The common base current gain, common emitter current gain and relationship between alpha and beta will always be confused. Inorder to avoid such confusion this activity is conducted.

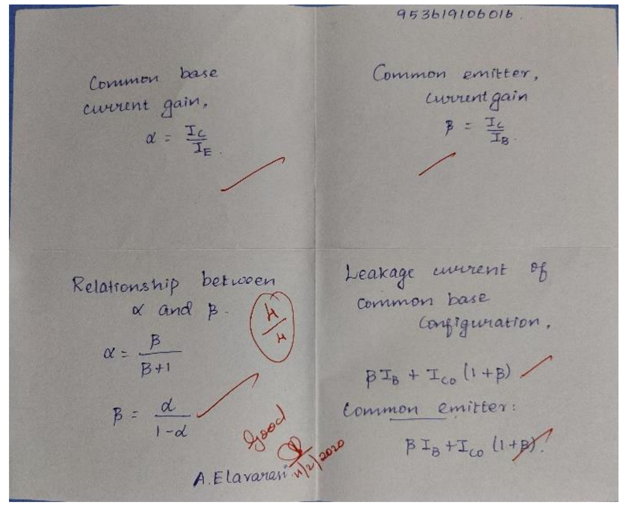
Time Allotted for the Activity: 5 Minutes

Details of the Implementation:

After teaching the concept of h parameter model, the students were given with a 4 fold paper. The Four corner four question were displayed in the power point presentation and students were asked to write the answer in each fold as shown below.







Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when most of the students wrote correctly.
- ✓ After completing the activity, the corrected paper with mistakes highlighted were given to the students.

Reflective Critique:

Benefits of conducting this activity:

- The concept of h parameter model can be easily applied for solving problems.
- It helps the student to perform modelling while designing amplifiers

Challenges:

- Few students got confused. Once the paper has been evaluated and given to the students understood the mistake done. The time for discussing the concept in the next class is more.

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	1	1	-	-	-	-	2	-	2	-	2

CO2: The students will be able to describe the basic geometry, operation and various configuration of Bipolar Junction Transistor.

References:

1. La Rudi, ‘Application of Teaching Model of Team Assisted Individualization (TAI) In Basic Chemistry Courses in Students of Forestry and Science of Environmental UniverstiasHalu Oleo’, International Journal of Education and Research,Vol. 5 No. 11 November 2017, pp 69-76.

2. Salivahanan S, Suresh Kumar N , Vallavaraj A, ‘ Electronic devices and circuits, 3 rd edition Tata Mcgraw Hill, 2008.